

Parkerian Hexad

(Six Elements of Information Security)

? What is Parkerian Hexad?

- It is an expanded model of information security that goes beyond the traditional CIA Triad.
- Introduced by Donn B. Parker
(Famous Security Researcher)
- Instead of just 3 elements (Confidentiality, Integrity, Availability), it defines 6 essential attributes of information security.

Why is it Important?

- CIA Triad is powerful, but sometimes too narrow for modern digital security challenges.
- The Parkerian Hexad adds three more attributes (Possession, Authenticity, Utility) which cover gaps left by CIA.
- It gives a more complete framework for analyzing, understanding, and designing security policies, systems, and defenses.

Confidentiality

- Ensures that information is accessible only to authorized individuals.
- Protects data from unauthorized disclosure.
- Encrypting your WhatsApp chats so only you and the receiver can read them.

Integrity

- Ensures accuracy, consistency, and trustworthiness of information.
- Prevents unauthorized modification of data.
- Online exam system ensuring that submitted answers cannot be altered by others.

Availability

- Ensures authorized users can access information and resources when needed.
- Protects against downtime and denial-of-service.
- Emergency ambulance system always available 24/7 without service interruption.

Possession (or Control)

- Refers to the ownership or control of information and physical media.
- Having access to data doesn't always mean having legal/authorized possession.
- If a hacker steals a USB with patient data, confidentiality might still be intact (if encrypted), but possession is lost.

Authenticity

- Verifies that information, transactions, and identities are genuine.
- Prevents impersonation and forgery.
- Digital signatures on emails confirm that the sender is authentic.

Utility

- Ensures that data is useful, meaningful, and in a usable format.
- Even if data is available, if it's corrupted or unusable, its utility is lost.
- A corrupted medical file that opens but shows unreadable text loses utility.